UC RIVERSIDE | Research and Economic Development

New UCR Research Seed Funding and Grant Program

Proposal title: Testing Significance of Moderator Effects on Multiple Subject Behavior Networks

NOTE: Preference will be given to applicants who have not previously received a SoCal OASIS® Internal Funding Award (OASIS-IFA). Previous recipients of OASIS-IFA may be considered provided they have submitted the final research and budget report for any prior award that has already expired.

Category of award applying for (please select one):

☐ Pre-seed Funding Award

Specify area ____________________

☒ Small Award

Specify area Fundamental Research

☐ Seed Funding Award

For Seed Funding Award please indicate a subcategory. Select one from below:

- ☐ Agriculture Technology and Food Security
- ☐ Health Related Research
- ☐ Human Development
- ☐ Natural Resource Management
- ☐ Renewable Energy and Fuels
- ☐ Sustainable Transportation and Infrastructure
- ☑ Fundamental Research and Other Areas not Covered by the Above

Total amount requested: $16,387 USD

Personnel

I. Principal Investigator:

Last Name: Sanchez Gomez First Name: Jose

Department: Statistics

School or College: CNAS

Phone: 919-697-6202 E-mail: josesa@ucr.edu

Present Title: Assistant Professor

II. CO-PIs: Names, titles, departments, colleges of all other personnel engaged on project. (Attach additional pages as needed.)

CO-PI: Last Name: O'Rourke First Name: Holly

Department: Psychology

School or College: CHASS

Phone: 602-317-1799 E-mail: holly.orourke@ucr.edu

CO-PI: Last Name: ____________ First Name: ____________

Department: ____________

School or College: ____________

Phone: ____________ E-mail: ____________

CO-PI: Last Name: ____________ First Name: ____________

Department: ____________

School or College: ____________

Phone: ____________ E-mail: ____________

CO-PI: Last Name: ____________ First Name: ____________

Department: ____________

School or College: ____________

Phone: ____________ E-mail: ____________

Submission materials:

1. This Application Form (pdf)
2. Research plan - No more than 2 pages for the Pre-seed Funding and Small Awards, single-spaced, and no more than 4 pages for the Seed Funding Awards. All narratives must use a 12-point font with at least one-inch margins on all sides. **Proposals not complying with these space and page limit requirements will not be reviewed.**
3. The narrative of the proposal should include a brief introduction , objectives, specific aims, and anticipated results (if applicable).
4. List of publications or creative works cited in the narrative, if any (no page limitations for this list).
5. Except for the Pre-seed funding awards, a budget outlining broad cost categories and a brief justification is required (no more than 2 pages for budget and justification). No F&A cost are allowed
6. CVs (no more than 2 pages for each investigator).
7. Results of prior SoCal OASIS IFA awards, if applicable (1 page max.).
8. **For the Seed funding awards** , external funding opportunities to be sought in the next 12 -18 months (1 page max.).

Testing Significance of Moderator Effects on Multiple Subject Behavior Networks

In psychological research, there is a pressing interest in understanding how socioeconomic status, physical health status, and prior mental health diagnoses may influence the emotional and behavioral patterns of individuals. A deeper understanding of these influences can aid in the development of comprehensive psychological profiles for subject assessment, and can ultimately inform personalized psychological interventions [1,2]. Motivated by these analytical needs, the goal of this project is to develop and make available a toolkit of novel statistical models and efficient algorithms to quantify and test the impact of between-person differences in the behavioral patterns of individuals over time.

This project is motivated by the investigators' collaborative project between the Departments of Psychology and Statistics at UCR [3], where we aim to use multi-subject longitudinal data to estimate Granger-causal networks that describe how depression, anxiety, stress, quality of sleep, and physical activity influence each other. An example of multi-subject Granger-causal networks is found in Figure 1. Nodes in the networks correspond to behavioral constructs, and links describe Granger-causal effects, *i.e.*, whether past values of one construct predict future values of another [4]. For example, a directed link from sleep quality to anxiety suggests that poor sleep quality will have a negative effect on next day's anxiety levels. Each of these links can potentially be affected by *moderators*, which are additional constructs that influence the strength of relations of the Granger-causal effects.

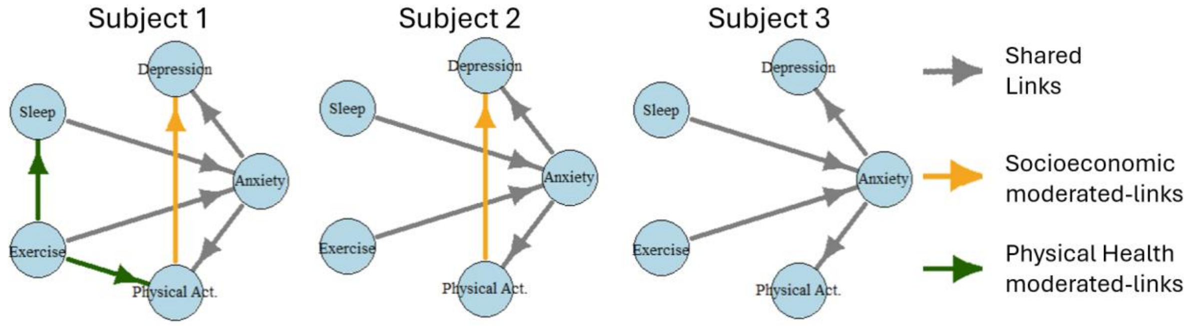

Figure 1: *Example of Granger-causal networks across 3 subjects. Each subject has a different behavioral network. Our aim is to quantify whether different network links are moderated by external socioeconomic status or health status.*

Prior methods in the literature can perform the estimation of such Granger-causal networks on multi-subject longitudinal data [5,6,7,8,9], and test network significance for a single subject [4,10,11,12]. However, these methods either do not incorporate moderator information [4,5,6,7,8,9,10,11,12] leading to false discoveries, or cannot test whether the estimated networks and moderator effects are statistically significant [5,6,7,8,9], leading to potentially biased results. The development of these tests is challenging because longitudinal observations are highly correlated, reducing the effective sample size [4]. Additionally, analyzing heterogeneous subjects increases model complexity, with the number of parameters growing combinatorially with the number of variables. For example, for 10 moderators, 10 variables in the Granger-causal network, and 50 subjects, we must assess a total of 22,500 parameters. Performing reliable inference in these high-dimensional settings with correlated observations is a substantial statistical and computational challenge.

In our collaborative project at UCR, we have developed the moderator vector autoregressive model (MOD-VAR), a new method that estimates individual Granger-causal networks and quantifies the effect of moderator variables on the network links of each subject [3]. MOD-VAR is effective in high-dimensional settings as confirmed via theoretical guarantees and extensive numerical experiments. This work is in preparation for submission to *Psychological Methods*.

Objectives

The core objective of this proposal is to develop a statistically rigorous hypothesis testing framework based on the MOD-VAR to assess, for each potential network edge and moderator variable, whether the moderator significantly influences Granger-causal effects across subjects. The effectiveness of our testing procedure will be verified through theoretical analysis, numerical experiments, and real-data applications. We aim to develop software implementation in R, making this toolkit widely accessible for analyzing multi-subject longitudinal data.
In the future, we aim to expand MOD-VAR beyond behavioral longitudinal studies through collaborations with interdisciplinary research teams at UCR and other institutions. We also plan to extend the theoretical idea to other statistical frameworks. Furthermore, we plan to apply for extramural funding to support the research activities of these extensions.

Specific aims

[A1] Develop the significance-testing procedure [Months 1-6]: Develop a multiple testing procedure controlling the false discovery rate across moderator-network edges, accounting for high dimensionality and subject heterogeneity.

[A2] Implement the method in R [Months 7-12]: Implement the testing method and release it as an R package.

[A3] Evaluate performance [Months 7-12]: Assess the testing procedure on simulated and real data. Apply the procedure to the GLOBEM dataset [13] as well as investigator-owned datasets to examine how socio-economic moderators influence behavioral patterns over time.

[A4] Explore extensions and external funding [Months 1-12]: Establish new interdisciplinary collaborations and prepare proposals for extramural funding.

Anticipated Results

We anticipate the development of the manuscript *"Testing Moderator Effects on Multiple-Subject Granger-Causal Networks on Intensive Longitudinal Data,"* together with a companion R package. In addition, we expect the establishment of new collaborative teams to extend MOD-VAR to general multi-subject longitudinal studies. One direction is to extend MOD-VAR for the analysis of brain connectivity networks using fMRI data to determine how moderators influence brain connectivity pathways [14,15,16]. A second direction is the analysis of multi-subject electroencephalogram (EEG) recordings to determine how brain activity varies across individuals [17,18,19,20,21]. Extramural funding will likely be required for the success of these innovative extensions. The UCR Research Small Grant will provide early support for the development of our MOD-VAR methodology and generate preliminary results vital for future grant applications and the long-term advancement of our research program.

References

[1] Molenaar, P. C. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement*, *2*(4), 201-218.

[2] Golino, H., Nesselroade, J., & Christensen, A. P. (2025). Toward a psychology of individuals: The ergodicity information index and a bottom-up approach for finding generalizations. *Multivariate Behavioral Research*, *60*(3), 528-555.

[3] Sánchez Gómez, J. A., O'Rourke, H, Brewer, G. (2026) Estimating Moderator Effects on Multiple Subject Intensive Longitudinal Studies. In preparation.

[4] Basu, S., & Michailidis, G. (2015). Regularized estimation in sparse high-dimensional time series models. *The Annals of Statistics, 43*(4) 1535-1567.

[5] Gates, K. M., & Molenaar, P. C. (2012). Group search algorithm recovers effective connectivity maps for individuals in homogeneous and heterogeneous samples. *NeuroImage*, *63*(1), 310-319.

[6] Gates, K. M., Lane, S. T., Varangis, E., Giovanello, K., & Guiskewicz, K. (2017). Unsupervised classification during time-series model building. *Multivariate Behavioral Research*, *52*(2), 129-148.

[7] Lane, S. T., Gates, K. M., Pike, H. K., Beltz, A. M., & Wright, A. G. (2019). Uncovering general, shared, and unique temporal patterns in ambulatory assessment data. *Psychological Methods*, *24*(1), 54.

[8] Fisher, Z. F., Kim, Y., Fredrickson, B. L., & Pipiras, V. (2022). Penalized estimation and forecasting of multiple subject intensive longitudinal data. *Psychometrika*, *87*(2), 403-431.

[9] Crawford, C. M., Park, J. J., Chow, S. M., Ernst, A. F., Pipiras, V., & Fisher, Z. F. (2024). Penalized subgrouping of heterogeneous time series. *arXiv preprint arXiv:2409.03085*.

[10] Krampe, J., Kreiss, J. P., Paparoditis, E. (2021) Bootstrap based inference for sparse high-dimensional time series models. *Bernoulli, 27*(3) 1441-1466.

[11] Zhu, K., & Liu, H. (2022). Confidence intervals for parameters in high-dimensional sparse vector autoregression. *Computational Statistics & Data Analysis*, *168*, 107383.

[12] Uematsu, Y., & Yamagata, T. (2025). Discovering the network Granger causality in large vector autoregressive models. *Journal of the American Statistical Association*, *120*(552), 2385-2396.

[13] Xu, X., Zhang, H., Sefidgar, Y., Ren, Y., Liu, X., Seo, W., ... & Dey, A. (2022). GLOBEM dataset: multi-year datasets for longitudinal human behavior modeling generalization. *Advances in Neural Information Processing Systems*, *35*, 24655-24692.

[14] Petersson, K. M., Nichols, T. E., Poline, J. B., & Holmes, A. P. (1999). Statistical limitations in functional neuroimaging II. Signal detection and statistical inference. *Philosophical transactions of the Royal Society of London. Series B: Biological Sciences*, *354*(1387), 1261-1281.

[15] Kherif, F., Poline, J. B., Mériaux, S., Benali, H., Flandin, G., & Brett, M. (2003). Group analysis in functional neuroimaging: selecting subjects using similarity measures. *Neuroimage*, *20*(4), 2197-2208.

[16] Ramsey, J. D., Hanson, S. J., & Glymour, C. (2011). Multi-subject search correctly identifies causal connections and most causal directions in the DCM models of the Smith et al. simulation study. *NeuroImage*, *58*(3), 838-848.

[17] Andrzejak, R. G., Lehnertz, K., Mormann, F., Rieke, C., David, P., & Elger, C. E. (2001). Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state. *Physical Review E*, *64*(6), 061907.

[18] Chaddad, A., Wu, Y., Kateb, R., & Bouridane, A. (2023). Electroencephalography signal processing: A comprehensive review and analysis of methods and techniques. *Sensors*, *23*(14), 6434.

[19] Fischer, M. H. F., Zibrandtsen, I. C., Høgh, P., & Musaeus, C. S. (2023). Systematic review of EEG coherence in Alzheimer’s disease. *Journal of Alzheimer’s Disease*, *91*(4), 1261-1272.

[20] Jiao, B., Li, R., Zhou, H., Qing, K., Liu, H., Pan, H., ... & Shen, L. (2023). Neural biomarker diagnosis and prediction to mild cognitive impairment and Alzheimer’s disease using EEG technology. *Alzheimer's research & therapy*, *15*(1), 32.

[21] Alvi, A. M., Siuly, S., & Wang, H. (2022). A long short-term memory-based framework for early detection of mild cognitive impairment from EEG signals. *IEEE Transactions on Emerging Topics in Computational Intelligence*, *7*(2), 375-388.

Budget Justification

Personnel - $10,887 USD.

To Be Named (TBN), Graduate Student (effort: 2 summer months).

Salary support for the GSR to assist in the development, implementation and evaluation of the MOD-VAR test.

- **GSR 1st month of 50% summer effort ($3,629 USD):** Develop, in coordination with the P.I., the method for testing moderator effects on behavioral networks, in accordance with Aim [A1].
- **GSR 2nd month of 50% summer effort ($3,629 USD):** Implement the MOD-VAR testing procedure in R (Aim [A2]).
- **GSR 3rd month of 50% summer effort ($3,629 USD):** Design simulations and conduct real data analyses on the HPCC cluster, based on previous simulation work (Aim [A3]).

Computational Resources – $3,500 USD.

- High Performance Computing Center (HPCC) subscription ($2,000 USD): Support for the subscriptions of the PI and co-PI's laboratories to the UCR HPCC for 12 months (June 2026 – July 2027), essential for Aims [A2, A3].
- GSR Computer Equipment ($1,500 USD): Purchase of computer equipment to enable GSR to work on Aims [A2, A3].

Coordination and Travel - $2,000 USD.

Funds to be allocated for PI to visit funding agencies, and for university visits to coordinate new collaborations (Aim [A4]).

Total Direct Costs – $16,387 USD.

Budget Summary

	Cost Element	Explanation	Total
1	**Personnel**		**$10,887 USD**
	GSR Summer Support	Funding for 3 summer months to support GSR research activities.	$10,887 USD
2	**Computational Resources**		**$3,500 USD**
	Computing Center Subscription	Funding to cover the annual subscription of the PI and co-PI's laboratory to the High-Performance Computing Center (HPCC) at UCR for the academic year of 2026-2027.	$2,000 USD
	Computer Resources for GSR	Funding to purchase computer equipment for the hired GSR.	$1,500 USD
3	**Coordination and Travel**		**$2,000 USD**
	Coordination and Travel	Cover the fee and travel costs to visit funding agencies in order to increase success rate of application to extramural funding and coordinate collaborations with interdisciplinary teams.	$2,000 USD
	Total	**(Direct costs)**	**$16,246 USD**

JOSE ANGEL SANCHEZ GOMEZ

Assistant Professor, Department of Statistics
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PROFESSIONAL EXPERIENCE

Assistant Professor August 2023 - current
Department of Statistics, University of California at Riverside (UCR) *Riverside, CA*

Data Science for the Social Good Fellow June-August 2022
German Research Center for Artificial Inteligence (DFKI) *Kaiserslautern, Germany*

EDUCATION

University of North Carolina at Chapel Hill, USA August 2018 - June 2023
Ph.D. in Statistics

The University of Guanajuato, Mexico August 2013 - June 2018
B.Sc. in Mathematics

RESEARCH MANUSCRIPTS

Peer-Reviewed Articles

- **Sánchez Gómez, J.**, Zhang, E., Liu, Y. (2025) *Effective Permutation Tests for Differences Across Multiple High-Dimensional Correlation Matrices.* Journal of Computational and Graphical Statistics, 1–10 https://doi.org/10.1080/10618600.2025.2550527.
- Cordero, J. M., Henn, T., Holtel, F., **Sánchez Gómez, J.**, Arenas, D., Sipka, A. Vollmer, S. (2025). *Developing Effective and Value-Aligned AI Tools for Journalists: 12 Critical Questions to Reflect upon.* Accepted in Journalism Practice https://doi.org/10.1080/17512786.2025.2465894.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. *Hub Detection in Gaussian Graphical Models.* Journal of the American Statistical Association 120(552), 2397-2409. https://doi.org/10.1080/01621459.2025.2453250.
- Griffith, A., **Sánchez Gómez, J.**, Castillo, K. (2024). *Hurricane Irma Linked to Coral Skeletal Density Shifts on The Florida Keys Reef Tract.* Integrative And Comparative Biology. https://doi.org/10.1093/icb/icae128
- Chowdhury, S., Clause, N., Memoli, F., **Sánchez Gómez, J.**, Wellner, Z. (2020) *New Families of Stable Simplicial Filtration Functors.* Topology and its Applications, 279(1), 107254. https://doi.org/10.1016/j.topol.2020.107254

In Preparation and In Progress

- **Sánchez Gómez, J.**, O'Rourke, H., Brewer, G. (2026) *Estimating Moderator Effects on Multiple Subject Intensive Longitudinal Studies.* In preparation since October 2024.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2026) *Joint Eigenspace Analysis: Perturbation Theory and Applications.* In progress since January, 2026.
- **Sánchez Gómez, J.**, Mo, W., Zhao, J., Liu, Y. (2026) *Identifying Common Hubs in Multiple Gaussian Graphical Models.* Under review at the 43rd International Conference on Machine Learning (ICML-2026).

- Cai, X., Gao, Q., **Sánchez Gómez, J.**, Au, K. (2026) *Cluster-assisted Gene Isoform Quantification: Theory and Practice.* In preparation since October 2023.
- Seongoh, P., **Sánchez Gómez, J.** (2026) *Transfer Learning for High-Dimensional Gaussian Graphical Models.* In preparation since September, 2024.
- **Sánchez Gómez, J.**, Whitney, A. (2026) *Machine Learning to Predict Local Government Deficits.* In preparation since October, 2021.

TEACHING EXPERIENCE

At the University of California at Riverside

- STAT 293: High-Dimensional Statistics (Fall 2025, Fall 2024, Fall 2023)
- STAT 107: Introduction to Statistical Computing with R (Fall 2025, Winter 2025, Winter 2024)
- STAT 155: Probability and Statistics for Science and Engineering (Winter 2024, Spring 2024)

PRESENTATIONS

- Invited Speaker "Identifying Common Hubs in Multiple Gaussian Graphical Models" at the Joint Statistical Meeting of the American Statistical Association in Nashville, Tennessee, August, 2025.
- Invited Speaker "Hub Detection in Gaussian Graphical Models" at the at Newport Beach, CA, November, 2024.
- Invited Speaker "Hub Detection in Gaussian Graphical Models" at the 7th International Conference on Econometrics and Statistics (EcoSta2024) at Beijing Normal University, July, 2024.
- Invited Speaker "Hub Detection in Gaussian Graphical Models" at the Institute of Mathematical Statistics Asia-Pacific Rim Meeting (IMS-APRM) at University of Melbourne, January, 2024.
- Presentation "Estimating hubs in Gaussian Graphical Models" in the Statistics Colloquium at the Ohio State University, February, 2023.
- Presentation "Estimating hubs in Gaussian Graphical Models" in the Statistics Colloquium at Southern Methodist University, January, 2023.
- Presentation "Estimating hubs in Gaussian Graphical Models" in the Statistics Colloquium at UC-Riverside, January, 2023.
- Presentation "Estimating hubs in Gaussian Graphical Models" in the Biomedical Research Facility at UC-San Diego, January, 2023.
- Presentation "Estimating hubs in Gaussian Graphical Models" in the Mathematics Colloquium at San Francisco State University, December, 2022.
- Presentation "Searching for New Topological Filtrations, Data" in the *Joint Meeting of the Mexican and Colombian Mathematical Societies* hosted in Barranquilla, Colombia, June, 2018.
- Poster "A Fırst Approach to a General Theory of Fıltratıon Functors" at the *2018 Latinx in the Mathematical Sciences Conference* at the Institute for Pure and Aplied Mathematics (IPAM), Los Angeles, USA, March, 2018.

HONORS AND AWARDS

2019 Cambanis-Hoeffding-Nicholson Award

For superior academic performance in the UNC-Chapel Hill doctoral program in Statistics.

HOLLY P. O'ROURKE, PHD

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CURRENT ACADEMIC APPOINTMENTS

2025–present: Associate Professor, Department of Psychology, University of California, Riverside
2025–present: Cooperating Faculty Member, Department of Statistics, University of California, Riverside

EDUCATION

PhD, 2016: Quantitative Psychology, Arizona State University
Dissertation: *Time Metric in Latent Difference Score Models*
MA, 2013: Quantitative Psychology, Arizona State University
Thesis: *Mediation as a novel method for increasing statistical power*
BS, 2009: Psychology, Arizona State University

RESEARCH INTERESTS

Quantitative methodology for behavioral science; longitudinal modeling; mediation analysis; latent change score models; zero-inflated count outcomes; machine learning for behavioral prediction; idiographic and nomothetic modeling; applications to health, prevention, education, and developmental science

SELECTED PEER-REVIEWED PUBLICATIONS

O'Rourke, H. P., & Hilley, C. D. (2025). A guide to specifying effects in latent change score models with moderated mediation. *Journal of Behavioral Data Science, 25*(2), 1–27.
O'Rourke, H. P., & Han, D. E. (2023). Considering the distributional form of zeroes when calculating mediation effects with zero-inflated count outcomes. *Journal of Behavioral Data Science, 3*(2), 1–14.
O'Rourke, H. P., Fine, K. L., Grimm, K. J., & MacKinnon, D. P. (2022). The importance of time metric precision when implementing bivariate latent change score models. *Multivariate Behavioral Research, 57*, 561–580.
Hilley, C. D., & **O'Rourke, H. P**. (2022). Dynamic change meets mechanisms of change: Mediation in the latent change score framework. *International Journal of Behavioral Development, 46*, 125–141.
Grimm, K. J., Helm, J., Rodgers, D., & **O'Rourke, H. P.** (2021). Analyzing cross-lag effects: A comparison of different cross-lag modeling approaches. *New Directions for Child and Adolescent Development, 175*, 11–33.
O'Rourke, H. P., & Vazquez, E. (2019). Mediation analysis with zero-inflated substance use outcomes: Challenges and recommendations. *Addictive Behaviors, 94*, 16–25.

SELECTED RESEARCH PRESENTATIONS

Brewer, G. A., & **O'Rourke, H. P.** (2026, July). *Exploring between-person and within-person variation in prospective memory*. Paper accepted at the 6th International Conference on Prospective Memory, Heidelberg, Germany.
O'Rourke, H. P. (2025, September). *From modeling linearity to machine learning: What's hot (and not) in the last century of data analysis in psychology*. Department of Psychology Annual Research Conference, UC Riverside.
O'Rourke, H. P. (2023, August). *Building superior multivariable systems to enhance precision in models of behavior change*. Presentation for Fralin Biomedical Research Institute at VTC, Virginia Tech, Roanoke, VA.
O'Rourke, H. P. (2020, September). *Mediation with zero-inflated count outcomes for substance use data:*

Negative binomial, Poisson, and hurdle models. McGill Quantitative Psychology Seminar, Montreal, Quebec, CA.

CURRENT EXTERNAL FUNDING

2023–2026: *Measuring Change in the Relations Amongst Cognitive, Health, and Socioemotional Factors at the Single-Subject Level and Improving Prediction of Critical Outcomes* (Army Research Institute / Army Materiel Command). Gene Brewer (PI), Holly O'Rourke (Co-PI). **$1,707,828** (49% recognition)

2024–2027: *Active Learning at Scale: Transforming Teaching and Learning via Large-Scale Learning Science and Generative AI* (Institute of Education Sciences). Danielle McNamara (PI), Arner, Brewer, Craig, Roscoe, Holly O'Rourke (Consultant). **$3,750,000** (5% recognition)

CURRENT INTERNAL FUNDING

2025–2026: *Improving Data Processing in Pursuit of Examining Intraindividual Variability in Cognition, Behavior, and Biological Health* (UCR CoR). O'Rourke (PI), Brewer (Co-I). **$4,000**

SELECTED COMPLETED FUNDING

2019–2021: *Using Computational Linguistics to Detect Comprehension Processes in Constructed Responses across Multiple Large Data Sets (Resubmission)* (DoEd). Danielle McNamara (PI), Holly O'Rourke (Co-PI), Renu Balyan (Co-PI). **$600,001** (10% recognition)

2019–2023 *iSTART-Early: Interactive Strategy Training for Active Reading and Thinking for Young Developing Readers (Resubmission)* (DoEd). Danielle McNamara (PI), Holly O'Rourke (Co-PI). **$1,400,000** (10% recognition)

2020–2022: *Multi-Component Behavioral Intervention Designed to Increase Functional Independence during Aging in ASD (Resubmission)* (DoD). B. Blair Braden (PI), Holly P. O'Rourke (Co-I). **$314,208** (30% recognition)

2018: *Interventions to Improve Divergent Thinking* (Laboratory for Analytic Sciences). Gene Brewer (PI), Holly O'Rourke (Co-PI), Adam Cohen (Co-I). **$84,991** (40% recognition)

TEACHING EXPERIENCE

UC Riverside
PSYC 259: Statistical Mediation Analysis (graduate)
PSYC 109: Advanced Research Methods and Statistics (undergraduate)

Arizona State University
FAS 598: Quantitative Methods in the Social Sciences I, FAS 508: Structural Equation Modeling (graduate)
SOC 390: Social Statistics I, FAS 498: Advanced Statistics for Social Sciences (undergraduate)

EDITORIAL AND REVIEW SERVICE

Editorial Board, Statistical Advisor: *BJPsych Open*
Review Editor, 2020–2025: *Frontiers in Psychology: Quantitative Psychology and Measurement*
Ad Hoc Reviewer: *Multivariate Behavioral Research, Structural Equation Modeling, Behavior Research Methods, Computational Brain & Behavior, Frontiers in Mathematics and Statistics, Addictive Behaviors, Prevention Science, Psychology of Addictive Behaviors, Child Development, Practical Assessment, Review, and Evaluation,* and others
Ad Hoc Grant Reviewer: National Science Foundation (NSF)

TECHNICAL EXPERTISE

Languages: R, SAS, Python, MATLAB, C/C++
Software: R, SAS, Mplus, MATLAB, Mathematica, SPSS
Systems: Windows, Macintosh, Unix, HPC/cloud computing